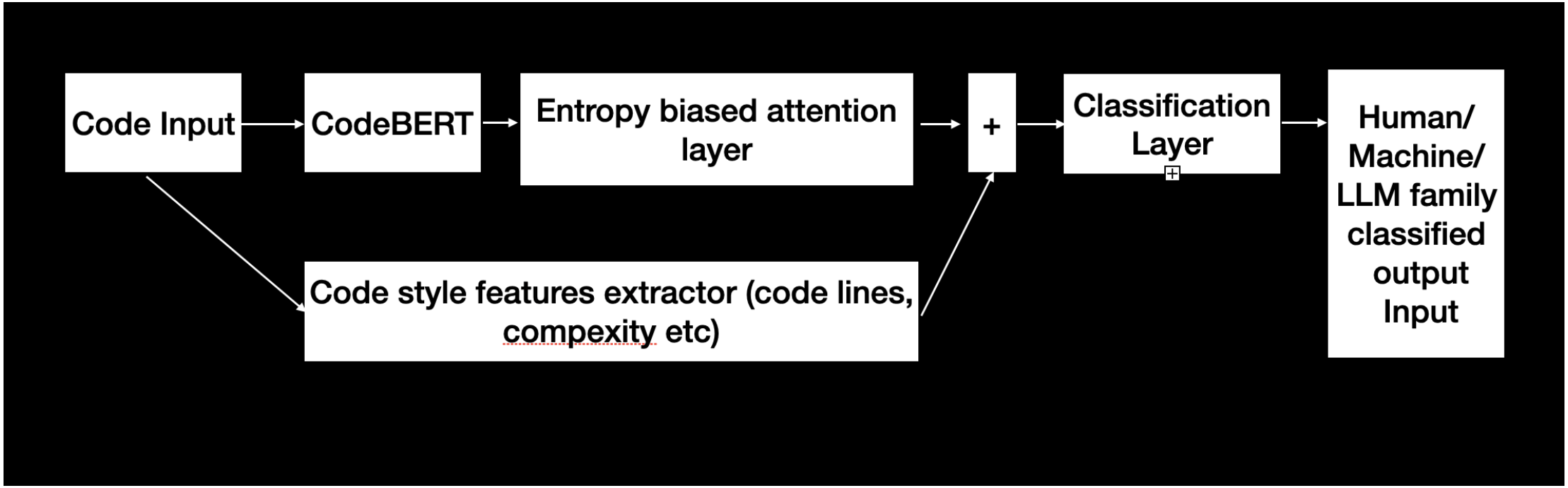
Machine Generated Code Detection

Team KVR

Architecture:



Tasks

- Task A:
- Human vs Machine generated code classification

- Task B:
- Machine generated code author(LLM) classification

Experimental Results – Task A

Model	Split	Samples	F1 (Macro)	Precision	Recall
Baseline	test_seenlang_seendom	37117	0.9731	0.9732	0.9730
	test_unseenlang_seendom	27722	0.5676	0.7525	0.6182
	test_seenlang_unseendom	46512	0.8692	0.8385	0.9086
	test_unseenlang_unseendom	349	0.5606	0.5451	0.6180
GPAD-1	test_seenlang_seendom	37117	0.9459	0.9459	0.9462
	test_unseenlang_seendom	27722	0.5194	0.6961	0.5824
	test_seenlang_unseendom	46512	0.8548	0.8453	0.8650
	test_unseenlang_unseendom	349	0.4898	0.4898	0.4898
GPAD-2	test_seenlang_seendom	37117	0.9957	0.9957	0.9957
	test_unseenlang_seendom	27722	0.7268	0.8106	0.7371
	test_seenlang_unseendom	46512	0.9579	0.9788	0.9392
	test_unseenlang_unseendom	17226	0.6538	0.8652	0.6411

Table 2: Updated evaluation metrics for Baseline, GPAD-1, and GPAD-2 across different train-test

Results Analysis – Task A

- Multi-Domain/Multi-Lingual Results:

- Performance on Seen Conditions: Highlighting GPAD-2's near-perfect 0.9957 F1 score
- Generalization to Unseen Languages: improvement over baseline
- Generalization to Unseen Domains: Strong 0.9579 F1 score

Experimental Results – Task B

Model Variant	Overall Acc.	Overall F1-M	Seen Acc.	Seen F1-M	Unseen Acc.	Unseen F1-M
CodeBERT Baseline	0.64	0.29	0.85	0.47	0.41	0.14
+ Entropy + Style Features	0.60	0.26	0.80	0.37	0.37	0.14
+ Style Only	0.62	0.20	0.76	0.19	0.46	0.12
+ Entropy Only	0.63	0.34	0.85	0.50	0.38	0.21

Table 3: Performance comparison of CodeBERT variants across seen and unseen author evaluation splits. F1-M = Macro F1.

Results Analysis – Task B

- Overall Performance Trends: Counterintuitive finding that Entropy Only performs best
- Seen vs. Unseen Gap: Large performance drop across all variants
- Entropy-Only Advantage: Relative improvement on unseen authors
- Style Feature Limitations: Why combined features underperform

Key Insights:

- GPAD-2 shows consistent improvements across all splits machine vs human detection
- Language shift is harder than domain shift
- Entropy features generalize better than style features for LLM author attribution
- Combination of features can sometimes hurt performance